IPTABLES

iptables is a powerful user-space command-line utility in Linux that configures IPv4 packet filtering rules within the Netfilter kernel framework. It acts as a firewall, managing incoming/outgoing network traffic, network address translation (NAT), and packet modification through defined tables, chains, and rules. Key commands include listing (-L), appending (-A), and flushing (-F) rules.

iptables is a user-space utility program that allows a system administrator to configure the IP packet filter rules of the Linux kernel firewall, implemented as different Netfilter modules.

Three components of IPTABLES : Tables , Chains, Rules

The filters are organized in a set of tables, which contain chains of rules for how to treat network traffic packets. Different kernel modules and programs are currently used for different protocols; *iptables* applies to IPv4, *ip6tables* to IPv6, *arptables* to ARP, and *ebtables* to Ethernet frames.

TABLES: 5 Tables : Filter Table, NAT Table, MANGLE Table,

Chains : Chains are like points in the route. There are 5 chains

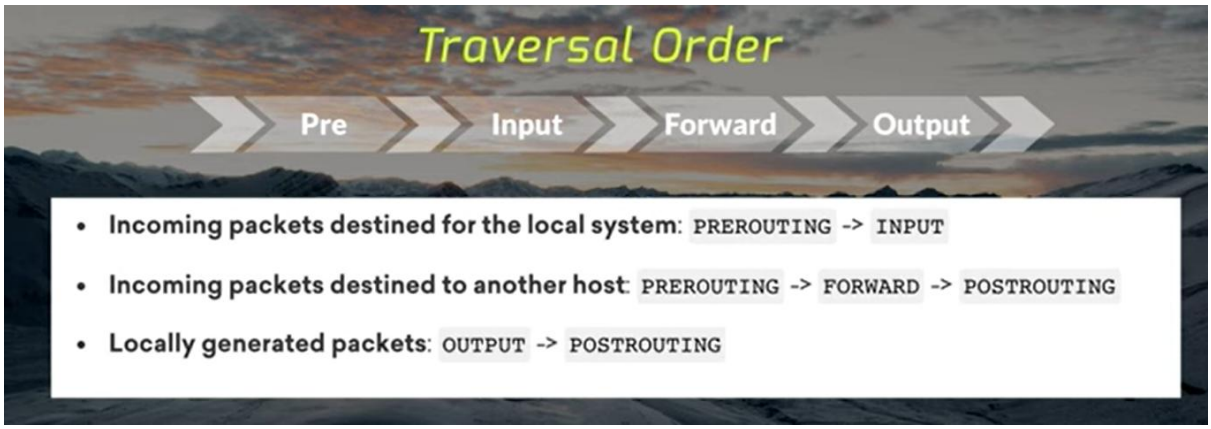
There are five *predefined chains* , though a table may not have all chains. Predefined chains have a *policy*, for example DROP, which is applied to the packet if it reaches the end of the chain. The system administrator can create as many other chains as desired. These chains have no policy; if a packet reaches the end of the chain it is returned to the chain which called it. A chain may be empty.

- **PREROUTING: Packets will enter this chain before a routing decision is made.**

- **INPUT: Packet is going to be locally delivered. It does not have anything to do with processes having an opened socket; local delivery is controlled by the "local-delivery" routing table: ip route show table local.**
- **FORWARD: All packets that have been routed and were not for local delivery will traverse this chain.**
- **OUTPUT: Packets sent from the machine itself will be visiting this chain.**
- **POSTROUTING: Routing decision has been made. Packets enter this chain just before handing them off to the hardware.**

A chain does not exist by itself; it belongs to a *table*. There are three tables: *nat*, *filter*, and *mangle*. Unless preceded by the option *-t*, an iptables command concerns the *filter* table by default. For example, the command iptables -L -v -n, which shows some chains and their rules, is equivalent to iptables -t filter -L -v -n. To show chains of table *nat*, use the command iptables -t nat -L -v -n

Tables↓/Chains→	PREROUTING	INPUT	FORWARD	OUTPUT	POSTROUTING
(routing decision)				✓	
raw	✓			✓	
(connection tracking enabled)	✓			✓	
mangle	✓	✓	✓	✓	✓
nat (DNAT)	✓			✓	
(routing decision)	✓			✓	
filter		✓	✓	✓	
security		✓	✓	✓	
nat (SNAT)		✓			✓



Rulues: Commands to manipulate network traffic

Rules 2 components : MATCHING COMPONENT , TARGET COMPONENT (ACTION) : terminating or non-terminating targets.

To block an IP Address : iptables -A INPUT -s 25.15.15.15 -j DROP

Terminating Targets : Accept, Drop, Queue, Reject, Return, User Defined

Non Terminating Targets: Chain continuation

EXERCISES

SYNTAX

Iptables -t [table] -OPTIONS[CHAIN] [matching component] [Action Component]

Options: A – Append, D – Delete, I – Insert, R – replace, Z – Zero counters, L – list, P – policy, E – Rename, F – Flush, X – Delete Chain

Matching component:

Generic: p- protocol, s-source IP, d-Destination IP, i-IN interface, 0-OUT interface

Implicit: TCP, -sport, -dport

Action Component:

-I (for jump)

ACCEPT, DROP, ReJECT, RETURN

To List an IPTABLE (default)

$ sudo iptables -L -n -v

```
xpstech@xpstech-PC: ~
xpstech@xpstech-PC:~$ sudo iptables -L -n -v
[sudo] password for xpstech:
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
xpstech@xpstech-PC:~$
```

To List an IPTABLE (mangle)

$ sudo iptables -t mangle -L -n -v

```
xpstech@xpstech-PC:~$ sudo iptables -t mangle -L -n -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
```

STEP 1: Check the Initial status

$ sudo iptables -L -n -v

```
xpstech@xpstech-PC: ~
xpstech@xpstech-PC:~$ sudo iptables -L -n -v
[sudo] password for xpstech:
Chain INPUT (policy ACCEPT 2473 packets, 3302K bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 1951 packets, 192K bytes)
 pkts bytes target     prot opt in     out     source               destination
xpstech@xpstech-PC:~$
```

No Rule Made Yet

STEP 2: Open a Web site an check whether it is opening or not

STEP 3 : Add rule to block a website

$ sudo iptables -A INPUT -s jntuh.ac.in -j DROP

STEP 4: Check whether rule is added or not

$ sudo iptables -L -n -v

```
xpstech@xpstech-PC:~$ sudo iptables -L -n -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 DROP       all  --  *      *       103.21.127.114       0.0.0.0/0

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 1 packets, 40 bytes)
 pkts bytes target     prot opt in     out     source               destination
xpstech@xpstech-PC:~$
```

STEP 5: Open a Web site an check whether it is opening or not

STEP 6: Delete the rule

$ sudo iptables -D INPUT 1

$ sudo iptables -L -n -v